

**National Coverage Determination (NCD)**

# PET Scans

**220.6**

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## Tracking Information

**Publication Number**

100-3

**Manual Section Number**

220.6

**Manual Section Title**

PET Scans

**Version Number****Effective Date of this Version**

01/28/2005

**Ending Effective Date of this Version**

04/03/2009

**Implementation Date**

04/18/2005

## Description Information

**Benefit Category**

Diagnostic Tests (other)

**Please Note:** This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

**Item/Service Description**

I. General Description

Positron emission tomography (PET) is a noninvasive diagnostic imaging procedure that assesses the level of metabolic activity and perfusion in various organ systems of the [human] body. A positron camera (tomograph) is used to produce cross-sectional tomographic images, which are obtained from positron emitting radioactive tracer substances (radiopharmaceuticals) such as 2-[F-18] Fluoro-D-Glucose (FDG), that are administered intravenously to the patient.

### Indications and Limitations of Coverage

The following indications may be covered for PET under certain circumstances. Details of Medicare PET coverage are discussed later in this section. Unless otherwise indicated, the clinical conditions below are covered when PET utilizes FDG as a tracer.

NOTE: This manual section 220.6 lists all Medicare-covered uses of PET scans. Except as set forth below in cancer indications listed as "Coverage with Evidence Development", a particular use of PET scans is not covered unless this manual specifically provides that such use is covered. Although this section 220.6 lists some non-covered uses of PET scans, it does not constitute an exhaustive list of all non-covered uses.

| Clinical Condition                | Effective Date  | Coverage                                                                |
|-----------------------------------|-----------------|-------------------------------------------------------------------------|
| Solitary Pulmonary Nodules (SPNs) | January 1, 1998 | Characterization                                                        |
| Lung Cancer (Non Small Cell)      | January 1, 1998 | Initial staging                                                         |
| Lung Cancer (Non Small Cell)      | July 1, 2001    | Diagnosis, staging, restaging                                           |
| Esophageal Cancer                 | July 1, 2001    | Diagnosis, staging, restaging                                           |
| Colorectal Cancer                 | July 1, 1999    | Determining location of tumors if rising CEA level suggests recurrence  |
| Colorectal Cancer                 | July 1, 2001    | Diagnosis, staging, restaging                                           |
| Lymphoma                          | July 1, 1999    | Staging and restaging only when used as alternative to Gallium scan     |
| Lymphoma                          | July 1, 2001    | Diagnosis, staging and restaging                                        |
| Melanoma                          | July 1, 1999    | Evaluating recurrence prior to surgery as alternative to Gallium scan   |
| Melanoma                          | July 1, 2001    | Diagnosis, staging, restaging; Non-covered for evaluating regional node |

|                                                   |                                    |                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Breast Cancer                                     | October 1, 2002                    | As an adjunct to standard imaging modalities for staging patients with distant metastasis or restaging patients with loco-regional recurrence or metastasis; as an adjunct to standard imaging modalities for monitoring tumor response to treatment for women with locally advanced and metastatic breast cancer when a change in therapy is anticipated |
| Head and Neck Cancers (excluding CNS and thyroid) | July 1, 2001                       | Diagnosis, staging, restaging                                                                                                                                                                                                                                                                                                                             |
| Thyroid Cancer                                    | October 1, 2003                    | Restaging of recurrent or residual thyroid cancers of follicular cell origin previously treated by thyroidectomy and radioiodine ablation and have a serum thyroglobulin >10ng/ml and negative I-131 whole body scan performed                                                                                                                            |
| Myocardial Viability                              | July 1, 2001 to September 30, 2002 | Only following inconclusive SPECT                                                                                                                                                                                                                                                                                                                         |
| Myocardial Viability                              | October 1, 2002                    | Primary or initial diagnosis, or following an inconclusive SPECT prior to revascularization. SPECT may not be used following an inconclusive PET scan                                                                                                                                                                                                     |
| Refractory Seizures                               | July 1, 2001                       | Pre-surgical evaluation only                                                                                                                                                                                                                                                                                                                              |
| Perfusion of the heart using Rubidium 82* tracer  | March 14, 1995                     | Noninvasive imaging of the perfusion of the heart                                                                                                                                                                                                                                                                                                         |
| Perfusion of the heart using ammonia N-13* tracer | October 1, 2003                    | Noninvasive imaging of the perfusion of the heart                                                                                                                                                                                                                                                                                                         |

\*Not FDG-PET.

EFFECTIVE JANUARY 28, 2005: This manual section lists Medicare-covered uses of PET scans effective for services performed on or after January 28, 2005. Except as set forth below in cancer indications listed as "coverage with evidence development", a particular use of PET scans is not covered unless this manual specifically provides that such use is covered. Although this section 220.6 lists some non-covered uses of PET scans, it does not constitute an exhaustive list of all non-covered uses.

For cancer indications listed as "coverage with evidence development" CMS determines that the evidence is sufficient to conclude that an FDG PET scan is reasonable and necessary only when the provider is participating in, and patients are enrolled in, one of the following types of prospective clinical studies that is designed to collect additional information at the time of the scan to assist in patient management:

- A clinical trial of FDG PET that meets the requirements of Food and Drug Administration (FDA) category B investigational device exemption (42 CFR 405.201);
- An FDG PET clinical study that is designed to collect additional information at the time of the scan to assist in patient management. Qualifying clinical studies must ensure that specific hypotheses are addressed; appropriate data elements are collected; hospitals and providers are qualified to provide the PET scan and interpret the results; participating hospitals and providers accurately report data on all enrolled patients not included in other qualifying trials through adequate auditing mechanisms; and, all patient confidentiality, privacy, and other Federal laws must be followed.

Effective January 28, 2005: For PET services identified as "Coverage with Evidence Development. Medicare shall notify providers and beneficiaries where these services can be accessed, as they become available, via the following:

- Federal Register Notice
- CMS coverage Web site at: [www.cms.gov/coverage](http://www.cms.gov/coverage)

| Indication                                     | Nationally<br>Covered <sup>1</sup> Non-<br>covered <sup>2</sup> | Coverage with Evidence<br>Development <sup>3</sup> |
|------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------|
| Brain                                          |                                                                 | X                                                  |
| Breast                                         |                                                                 |                                                    |
| -Diagnosis                                     |                                                                 | X                                                  |
| -Initial staging of axillary nodes             |                                                                 | X                                                  |
| -Staging of distant metastasis                 | X                                                               |                                                    |
| -Restaging, monitoring *                       | X                                                               |                                                    |
| Cervical                                       |                                                                 |                                                    |
| -Staging as adjunct to conventional<br>imaging | X                                                               |                                                    |
| -Other staging                                 |                                                                 | X                                                  |
| -Diagnosis, restaging, monitoring *            |                                                                 | X                                                  |
| Colorectal                                     |                                                                 |                                                    |
| -Diagnosis, staging, restaging                 | X                                                               |                                                    |
| -Monitoring *                                  |                                                                 | X                                                  |
| Esophagus                                      | X                                                               |                                                    |
| -Diagnosis, staging, restaging                 |                                                                 |                                                    |

|                                                          |   |   |
|----------------------------------------------------------|---|---|
| -Monitoring *                                            |   | X |
| Head and Neck (non-CNS/thyroid)                          |   |   |
| -Diagnosis, staging, restaging                           | X |   |
| -Monitoring *                                            |   | X |
| Lymphoma                                                 |   |   |
| -Diagnosis, staging, restaging                           | X |   |
| -Monitoring *                                            |   | X |
| Melanoma                                                 |   |   |
| -Diagnosis, staging, restaging                           | X |   |
| -Monitoring *                                            |   | X |
| Non-Small Cell Lung                                      |   |   |
| -Diagnosis, staging, restaging                           | X |   |
| -Monitoring *                                            |   | X |
| Ovarian                                                  |   | X |
| Pancreatic                                               |   | X |
| Small Cell Lung                                          |   | X |
| Soft Tissue Sarcoma                                      |   | X |
| Solitary Pulmonary Nodule<br>(characterization)          | X |   |
| Thyroid                                                  |   |   |
| -Staging of follicular cell tumors                       | X |   |
| -Restaging of medullary cell tumors                      |   | X |
| -Diagnosis, other staging & restaging                    |   | X |
| -Monitoring *                                            |   | X |
| Testicular                                               |   | X |
| All other cancers not listed herein (all<br>indications) |   | X |

<sup>1</sup> Covered nationally based on evidence of benefit. Refer to National Coverage Determination Manual Section 220.6 in its entirety for specific coverage language and limitations for each indication.

<sup>2</sup> Non-covered nationally based on evidence of harm or no benefit.

<sup>3</sup> Covered only in specific settings discussed above if certain patient safeguards are provided. Otherwise, non-covered nationally based on lack of evidence sufficient to establish either benefit or harm or no prior decision addressing this cancer. Medicare shall notify providers and beneficiaries where these services can be accessed, as they become available, via the following:

- Federal Register Notice

- CMS coverage Web site at: [www.cms.gov/coverage](http://www.cms.gov/coverage)

\* Monitoring = monitoring response to treatment when a change in therapy is anticipated.

## II. General Conditions of Coverage for FDG PET

### Allowable FDG PET Systems

#### A. Definitions: For purposes of this section:

- "Any FDA-approved" means all systems approved or cleared for marketing by the Food and Drug Administration (FDA) to image radionuclides in the body.
- "FDA-approved" means that the system indicated has been approved or cleared for marketing by the FDA to image radionuclides in the body.
- "Certain coincidence systems" refers to the systems that have all the following features:
  - Crystal at least 5/8-inch thick;
  - Techniques to minimize or correct for scatter and/or randoms; and
  - Digital detectors and iterative reconstruction.

Scans performed with gamma camera PET systems with crystals thinner than 5/8" will not be covered by Medicare. In addition, scans performed with systems with crystals greater than or equal to 5/8" in thickness, but that do not meet the other listed design characteristics are not covered by Medicare.

#### B. Allowable PET systems by covered clinical indication:

| Covered Clinical Condition                                                        | Allowable Type of FDG PET System     |                                                 |                                                              |
|-----------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------|--------------------------------------------------------------|
|                                                                                   | Prior to<br>July 1, 2001             | July 1, 2001<br>through<br>December 31,<br>2001 | On or after<br>January 1, 2002                               |
| Characterization of single pulmonary nodules                                      | Effective 1/1/1998, any FDA-approved | Any FDA-approved                                | FDA-approved: Full/Partial ring, certain coincidence systems |
| Initial staging of lung cancer (non small cell)                                   | Effective 1/1/1998, any FDA-approved | Any FDA-approved                                | FDA-approved: Full/Partial ring, certain coincidence systems |
| Determining location of colorectal tumors if rising CEA level suggests recurrence | Effective 7/1/1999, any FDA-approved | Any FDA-approved                                | FDA approved: Full/Partial ring, certain coincidence systems |
| Staging or restaging of lymphoma only when used as alternative to gallium scan    | Effective 7/1/1999, any FDA-approved | Any FDA-approved                                | FDA-approved: Full/Partial ring, certain coincidence systems |

|                                                                                       |                                       |                  |                                                              |
|---------------------------------------------------------------------------------------|---------------------------------------|------------------|--------------------------------------------------------------|
| Evaluating recurrence of melanoma prior to surgery as alternative to gallium scan     | Effective 7/1/1999, any FDA-approved. | Any FDA-approved | FDA-approved: Full/Partial ring, certain coincidence systems |
| Diagnosis, staging, restaging of colorectal cancer                                    | Not covered by Medicare               | Full ring        | FDA-approved: Full/Partial ring                              |
| Diagnosis, staging, restaging of esophageal cancer                                    | Not covered by Medicare               | Full ring        | FDA-approved: Full/Partial ring                              |
| Diagnosis, staging, restaging of head and neck cancers (excluding CNS and thyroid)    | Not covered by Medicare               | Full ring        | FDA-approved: Full/Partial ring                              |
| Diagnosis, staging, restaging of lung cancer (non small cell)                         | Not covered by Medicare               | Full ring        | FDA-approved: Full/Partial ring                              |
| Diagnosis, staging, restaging of lymphoma                                             | Not covered by Medicare               | Full ring        | FDA-approved: Full/Partial ring                              |
| Diagnosis, staging, restaging of melanoma (non-covered for evaluating regional nodes) | Not covered by Medicare               | Full ring        | FDA-approved: Full/Partial ring                              |
| Determination of myocardial viability only following inconclusive SPECT               | Not covered by Medicare               | Full ring        | FDA-approved: Full/Partial ring                              |
| Pre-surgical evaluation of refractory seizures                                        | Not covered by Medicare               | Full ring        | FDA-approved: Full ring                                      |
| Breast Cancer                                                                         | Not covered                           | Not covered      | Effective October 1, 2002, Full/Partial ring                 |
| Thyroid Cancer                                                                        | Not covered                           | Not covered      | Effective October 1, 2003, Full/Partial ring                 |
| Myocardial Viability Primary or initial diagnosis prior to revascularization          | Not covered                           | Not covered      | Effective October 1, 2002, Full/Partial ring                 |
| All other oncology indications not previously specified                               | Not covered                           | Not covered      | Effective January 28, 2005, Full/Partial ring                |

C. Regardless of any other terms or conditions, all uses of FDG PET scans, in order to be covered by the Medicare program, must meet the following general conditions prior to June 30, 2001:

- Submission of claims for payment must include any information Medicare requires to ensure the PET scans performed were: (a) medically necessary, (b) did not unnecessarily duplicate other covered diagnostic tests, and (c) did not involve investigational drugs or procedures using investigational drugs, as determined by the FDA.

- The PET scan entity submitting claims for payment must keep such patient records as Medicare requires on file for each patient for whom a PET scan claim is made.

Regardless of any other terms or conditions, all uses of FDG PET scans, in order to be covered by the Medicare program, must meet the following general conditions as of July 1, 2001:

- The provider of the PET scan should maintain on file the doctor's referral and documentation that the procedure involved only FDA-approved drugs and devices, as is normal business practice.
- The ordering physician is responsible for documenting the medical necessity of the study and ensuring that it meets the conditions specified in the instructions. The physician should have documentation in the beneficiary's medical record to support the referral to the PET scan provider.

### III. Covered Indications for PET Scans and Limitations/Requirements for Usage

For all uses of PET relating to malignancies the following conditions apply:

A. Diagnosis: PET is covered only in clinical situations in which: (1) the PET results may assist in avoiding an invasive diagnostic procedure, or in which (2) the PET results may assist in determining the optimal anatomical location to perform an invasive diagnostic procedure. In general, for most solid tumors, a tissue diagnosis is made prior to the performance of PET scanning. PET scans following a tissue diagnosis are generally performed for staging rather than diagnosis.

PET is not covered as a screening test (i.e., testing patients without specific signs and symptoms of disease).

B. Staging: PET is covered for staging in clinical situations in which: (1)(a) the stage of the cancer remains in doubt after completion of a standard diagnostic workup, including conventional imaging (computed tomography (CT), magnetic resonance imaging (MRI), or ultrasound), or (1)(b) it could potentially replace one or more conventional imaging studies when it is expected that conventional study information is insufficient for the clinical management of the patient, and 2) clinical management of the patient would differ depending on the stage of the cancer identified.

C. Restaging: PET is covered for restaging: (1) after completion of treatment for the purpose of detecting residual disease, (2) for detecting suspected recurrence or metastasis, (3) to determine the extent of a known recurrence, or (4) if it could potentially replace one or more conventional imaging studies when it is expected that conventional study information is insufficient for the clinical management of the patient. Restaging applies to testing after a course of treatment is completed, and is covered subject to the conditions above.



D. Monitoring: This refers to use of PET to monitor tumor response to treatment during the planned course of therapy (i.e., when a change in therapy is anticipated ).

NOTE: In the absence of national frequency limitations, contractors, should, if necessary, develop frequency requirements on any or all of the indications covered on and after July 1, 2001.

(This NCD last reviewed December 2004.)

### Cross Reference

See NCD for [SPECT](#) (§220.12).

### Claims Processing Instructions

[TN 527 \(Medicare Claims Processing Manual\)](#)

[TN 628 \(Medicare Claims Processing Manual\)](#)

[TN 1817 \(Medicare Claims Processing\)](#)

[TN 1833 \(Medicare Claims Processing\)](#)

[TN 2750 \(Medicare Claims Processing\)](#)

[TN 11272 \(Medicare Claims Processing\)](#)

[TN 11426 \(Medicare Claims Processing\)](#)

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## Transmittal Information

### Transmittal Number

31

### Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R31NCD.pdf>

### Revision History

04/2005 - Covered as an adjunct test for detection of pre-treatment metastases (i.e., staging) in newly diagnosed cervical cancer subsequent to conventional imaging that is negative for extra-pelvic metastasis. For all remaining indications that are subject of this NCD, as well as other indications for cervical cancer not specifically mentioned, covered only when providers are participating in, and patients are enrolled in, an approved FDG PET clinical study, or an FDG PET clinical trial meeting FDA category B IDE exemption status. Effective date 01/28/2005. Implementation date 4/18/2005. ([TN 31](#)) (CR 3741)

06/2003 - Expanded coverage for noninvasive imaging of perfusion of heart using FDA-approved Ammonia N-13 tracer, and restaging of recurrent or residual thyroid cancers of follicular cell origin that have been previously treated by thyroidectomy and radioiodine ablation and have a serum thyroglobulin >10ng/ml and negative I-131 whole body scan. Also maintained national noncoverage determinations for all uses of FDG-PET for soft tissue sarcoma and dementia and neurogenerative diseases. Effective and implementation dates 10/01/2003. ([TN 171](#)) (CR 2687)

05/2002 - Provided coverage for breast cancer and broader coverage for myocardial viability. Effective and implementation dates 10/01/2002. ([TN 156](#)) (CR 2138)

11/2001 - Clarified types of allowable PET scanners. Effective and implementation dates 01/01/2002. ([TN 147](#)) (CR 1886) Reduction in coverage announced in Federal Register, 66 FR 58741.

04/2001 - Expanded coverage to include diagnosis and staging/restaging of non-small cell lung cancer, esophageal cancer, colorectal cancer, lymphoma, melanoma, head and neck cancers, myocardial viability, and refractory seizures. Effective and implementation dates July 1, 2001. ([TN 136](#)) (CR 1603)

04/1999 - Added coverage for evaluation of recurrent colorectal cancer in patients with rising levels of carcinoembryonic antigen, staging of lymphoma (both Hodgkins and non-Hodgkins) when PET scan substitutes for gallium scan, and detection of recurrent melanoma. Effective July 1, 1999. (TN 113) (CR 861)

02/1999 - Added coverage for characterization of solitary pulmonary nodules and initial staging of lung cancer. Effective date 1/1/1998. (TN 106)

05/1995 - Provided limited coverage for use in noninvasive imaging of perfusion of heart for diagnosis and management of patients with known or suspected coronary artery disease. Effective date 03/14/1995. (TN 76)

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## National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Positron Emission Tomography \(FDG\) \(CAG-00065N\)](#)
- [First reconsideration for Positron Emission Tomography \(NaF-18\) to Identify Bone Metastasis of Cancer \(CAG-00065R\)](#)

- [Original Consideration for Positron Emission Tomography \(FDG\) for Alzheimer's Disease/Dementia \(CAG-00088N\)](#) 
- [First reconsideration for Positron Emission Tomography \(FDG\) and Other Neuroimaging Devices for Suspected Dementia \(CAG-00088R\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Breast Cancer \(CAG-00094N\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Brain, Cervical, Ovarian, Pancreatic, Small Cell Lung, and Testicular Cancers \(CAG-00181N\)](#) 
- [First reconsideration for Positron Emission Tomography \(FDG\) for Solid Tumors \(CAG-00181R\)](#) 
- [Second reconsideration for Positron Emission Tomography \(FDG\) for Cervical Cancer \(CAG-00181R2\)](#) 
- [Original Consideration for Positron Emission Tomography \(FDG\) for Infection and Inflammation \(CAG-00382N\)](#) 

## Additional Information

### Other Versions

| Title                                              | Version | Effective Between       |  |
|----------------------------------------------------|---------|-------------------------|---------------------------------------------------------------------------------------|
| Positron Emission Tomography (PET) Scans - RETIRED | 7       | 04/10/2023 - N/A        | <a href="#">View</a>                                                                  |
| Positron Emission Tomography (PET) Scans           | 6       | 01/01/2022 - 04/10/2023 | <a href="#">View</a>                                                                  |
| Positron Emission Tomography (PET) Scans           | 5       | 03/07/2013 - 01/01/2022 | <a href="#">View</a>                                                                  |
| Positron Emission Tomography (PET) Scans           | 4       | 04/03/2009 - 03/07/2013 | <a href="#">View</a>                                                                  |

| Title     | Version | Effective Between          |  |
|-----------|---------|----------------------------|-------------------------------------------------------------------------------------|
| PET Scans | 3       | 01/28/2005 -<br>04/03/2009 | You are<br>here                                                                     |
| PET Scans | 2       | 10/01/2003 -<br>01/28/2005 | <a href="#">View</a>                                                                |
| PET Scans | 1       | 10/01/2002 -<br>10/01/2003 | <a href="#">View</a>                                                                |